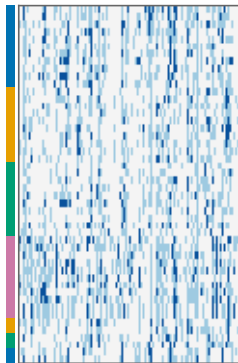


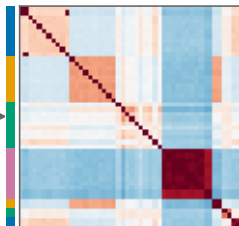
aarobust-kin: robust PCA (PCP) of the GRM with a kinship threshold

genotypes G



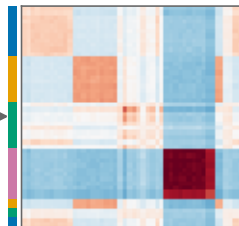
standardise

GRM C
(centred, scaled)



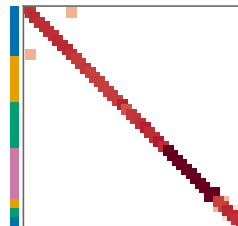
PCP
 $C = L + S$

L : diagonal and
pairs imputed



+

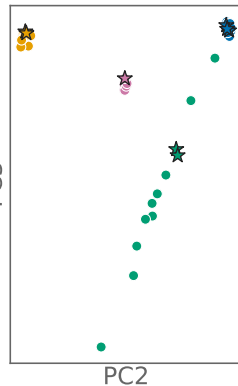
S : diagonal
+ related pairs



eigen-
vectors

PC3

PCs of L



CEU



CHB



MXL



YRI



relative